

Regular Grammars

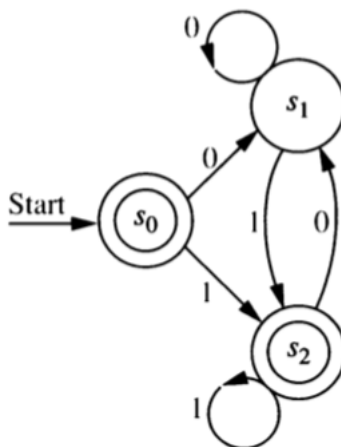
Math 252

Grammars to Automata

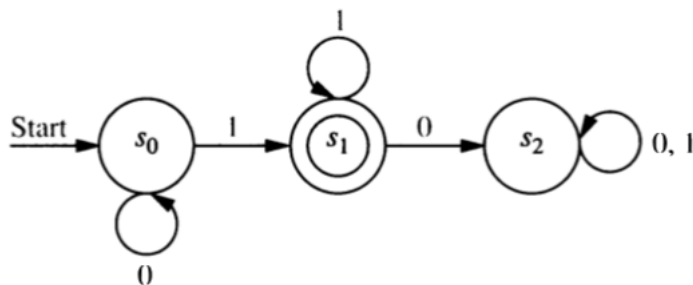
1. Construct a finite state automaton that recognizes that language generated by the following regular grammar.
 - terminals: 0 and 1
 - nonterminals: S and A
 - start symbol: S .
 - production rules: $S \rightarrow 1A$, $S \rightarrow 0$, $S \rightarrow \lambda$, $A \rightarrow 0A$, $A \rightarrow 1A$, $A \rightarrow 1$
2. Construct a finite state automaton that recognizes that language generated by the following regular grammar.
 - terminals: 0 and 1
 - nonterminals: S , A , B
 - start symbol: S .
 - production rules: $S \rightarrow 1A$, $S \rightarrow 0$, $A \rightarrow 0B$, $A \rightarrow 1A$, $A \rightarrow 1 B \rightarrow 0A$, $B \rightarrow 1B$, $B \rightarrow 0$
3. Explain how *any* regular grammar G can be converted into an NFA N such that $L(N) = L(G)$.
 - a. What is a regular grammar? (What sorts of production rules are allowed?)
 - b. What will the states of N be? How many will there be?
 - c. Which states will be final (ie. accepting) states?
 - d. How does each rule in G get converted into a part of N ? [Go through each kind of rule a regular grammar may have.]

Automata to Grammars

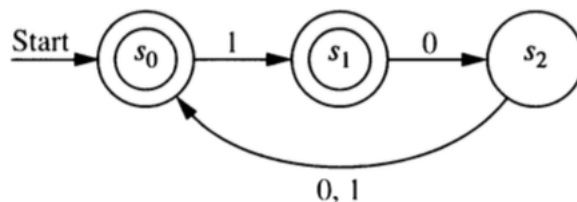
4. Construct a regular grammar that generates the language accepted by this finite state automaton.



5. Construct a regular grammar that generates the language accepted by this finite state automaton.



6. Construct a regular grammar that generates the language accepted by this finite state automaton.



7. Explain how we know that *any* regular language L is generated by a regular grammar G by describing how to construct the regular grammar G .
- We already know that every regular expression can be converted into an NFA and every NFA into a DFA. So all we need to show is that we can convert DFAs or NFAs into regular grammars. Are there any advantages to starting from a DFA over an NFA?
 - What will the terminals and nonterminals in your grammar be?
 - What will the start symbol be?
 - How are the production rules created?
8. Show that every DFA is equivalent to a DFA with a “lonely start state”. A lonely start state is a start state that can never be returned to. (In terms of the graph, its in-degree is 0.)